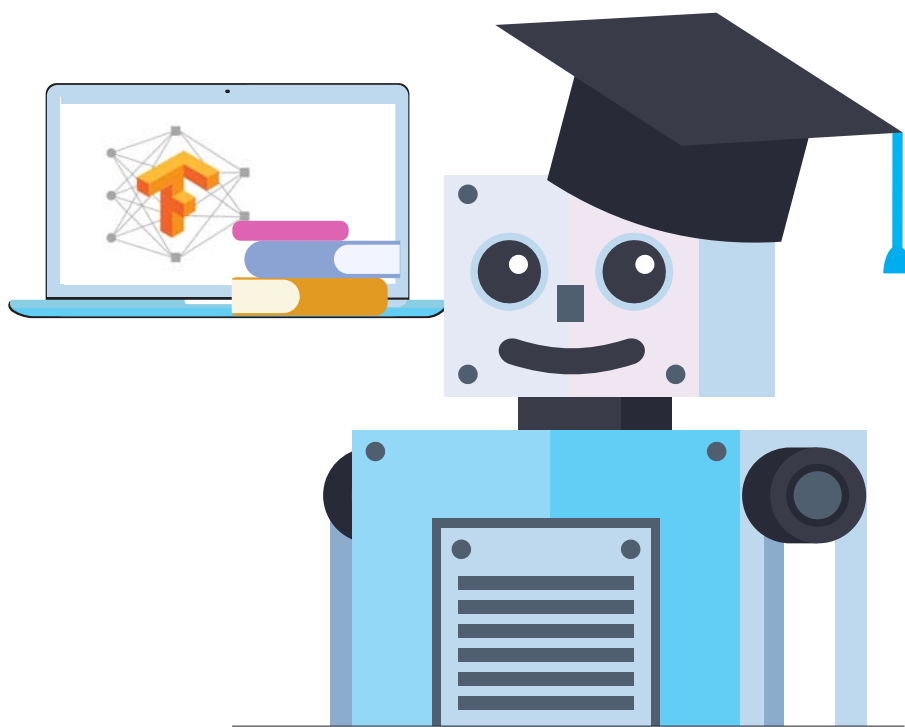


AI: An Introduction to Probability

In the previous article in this series on AI (August 2022 issue of *OSFY*), we continued our discussion of matrices and linear algebra. We also discussed a few Python libraries useful in developing AI based applications, and learned to use JupyterLab to run our Python code. In this fourth article in the series on AI, we will begin exploring TensorFlow, a very powerful library used for developing AI and machine learning applications. We will also briefly discuss a few other useful libraries on the way. Later, we will discuss probability, theory as well as code. But as always, we will start by discussing a few topics that will widen our understanding of AI.



alone. This definitely tells us that in the long run we need to develop AI based applications with accountability and some guarantees, not just some results taken out of a magic box. Currently, a lot of research is trying to answer these questions.

The deployment of AI based applications also raises a lot of moral and ethical questions. Moreover, you don't need to wait till the emergence of strong AI (also known as artificial general intelligence) to study its social impact. As an example, imagine you are driving a car through a road with hairpin bends on a rainy night. Assume that the car is running at a speed of around 100 km per hour. Suddenly, someone crosses the road. What will your response be? Since this is a thought experiment, let us make the situation even more complicated. If you suddenly apply the brakes or turn the car, your life will be in great danger. But if you don't do this the life of the person crossing the road will be in jeopardy. For us humans, this is a split second decision and even the most selfish person may decide to save the pedestrian, because we have the trait of self-sacrifice. But how do we ever teach an AI based system to imitate this

So far, we have discussed only the technological aspects of AI. What about the social aspects? Maybe, it is time to analyse the social implications of deploying more and more AI based applications. Imagine a scenario where you are applying for a job, which is processed by a group of people. They go through your resume and make a decision. In case you are not selected for the job, you can get feedback from the group that processed

your application. They may even be able to convince you about the reasons for rejecting your application. Now, assume that your application is being processed by software powered with AI. Again, your application may get rejected. However, this time there is no accountability. You cannot ask an AI based software system for a feedback. So, in this scenario you are not sure whether the rejection of your application was indeed based on merit